

Logic Gates — Practice · AS 9618 — ANSWERS (student copy) · first edition, the sheet set as homework

1. $X = (\text{NOT } A) \text{ AND } (B \text{ OR } C)$. X column, rows 000 $\rightarrow$ 111: 0, 1, 1, 1, 0, 0, 0, 0 — 1 mark per 2 correct rows
2. $X = \text{NOT } (A \text{ AND } B) \text{ AND } (B \text{ OR } C)$ — accept $(A \text{ NAND } B) \text{ AND } (B \text{ OR } C)$. Marks: NOT(A AND B) [1], (B OR C) [1], both combined with AND [1]
3. (a) **NAND** (b) **NOR**
4. **XOR** (output 1 only when inputs differ)
5. (a) $2^4 = 16$ (b) 2^n
6. **A = 1, B = 1** (OR $\rightarrow$ 1, XOR $\rightarrow$ 0; on the other three rows they agree)

Total: 13 marks. Q2 trap: dropping the NOT on the NAND branch, or attaching NOT to only one input — NOT (A AND B) $\neq$ (NOT A) AND B.

 **The quiet one, unhinted.** Question 3(a) is NAND and 3(b) is NOR — and those two are **the only gates in existence from which every other gate can be built**. They are called the universal gates, and the sheet has them sitting as (a) and (b) of one question as if they were any other pair.